Non-Injectivity as a Structural Necessity of Genuine Emergence

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Abstract

We prove that any genuinely emergent physical description is necessarily non-injective and intrinsically compressive. The argument is purely structural: given a surjective projection $\Pi : \Omega \to \mathcal{O}$ whose observables are entirely defined by Π, if the effective level is not structurally isomorphic to the fundamental level, then Π cannot be injective, and the associated projection entropy $S_\Pi > 0$. The proof uses only surjectivity, the definition of observables by projection, and the non-isomorphism condition; it is independent of any specific physical framework. As a corollary, descriptive redundancy, non-injectivity of Π, and $S_\Pi > 0$ are co-extensive in any projective emergent framework, and any non-trivial gauge structure requires these properties. The result reframes non-injectivity from a modelling choice to a structural necessity: a fully injective emergent description would reduce to a mere reparametrisation of the fundamental level. We discuss how this universal constraint is explicitly realised in the Cosmochrony relational framework [1], where non-injectivity of the projection Π underlies the emergence of gauge structure, spacetime geometry, and quantum correlations [5, 2].

Keywords. emergence, non-injectivity, projection, gauge symmetry, information loss, entropy, effective theories.

1 Introduction

Many physical frameworks assume the existence of two distinct levels of description: a fundamental level Ω of underlying configurations, and an effective level $\mathcal{O}$ of observable states. The relationship between these levels is typically encoded in a projection map $\Pi : \Omega \to \mathcal{O}$.

A recurring feature of such frameworks is that Π is many-to-one: distinct configurations $\omega \in \Omega$ may correspond to the same observable state $o \in \mathcal{O}$. This non-injectivity has been identified as the structural origin of gauge symmetry [1], Bell-inequality violations [2], and black hole entropy [3]. In each case, however, non-injectivity has been introduced as a feature of a specific model rather than derived as a universal necessity.

The present note closes this logical gap. We show that non-injectivity is not a modelling choice but a structural consequence of genuine emergence: any surjective map Π whose observables are entirely defined by Π, and whose effective level is not isomorphic to the fundamental level, must be non-injective. The proof is elementary and framework-independent.

A preliminary observation: standard cosmology is emergent without saying so

Before stating our results, we draw attention to a structural feature of any Big Bang cosmology that motivates the theorem.

In the standard cosmological model ΛCDM, an effective spacetime geometry $g_{\mu\nu}$ is well defined and descriptively valid for $t > t_{\text{Planck}}$. For $t \lesssim t_{\text{Planck}}$, the geometric description ceases to be valid: the metric diverges, the Einstein equations break down, and $g_{\mu\nu}$ is not a well-defined object. Yet the large-scale geometry that exists for $t \gg t_{\text{Planck}}$ is stable, continuous, and physically meaningful.

These three facts jointly imply that $g_{\mu\nu}$ is not fundamental. A fundamental structure does not cease to exist in any regime; it is the entity from which descriptions in all regimes are derived. Since $g_{\mu\nu}$ exists for $t > t_{\text{Planck}}$ but not before, it must be produced by a more primitive structure Ω whose projection Π gives rise to it when conditions for projectability are met. In other words: any theory that admits a regime where its own description ceases to be valid is already a theory of emergence — but an incomplete one, in that it has suppressed the specification of Ω and Π.

> *Any cosmological framework admitting a regime where spacetime geometry ceases to be well-defined is, by necessity, an emergent description: it implicitly assumes the existence of a more fundamental level from which geometry arises. The absence of an explicit formulation of this underlying level does not remove the emergence — it only renders the theory incomplete.*

Saying "we do not know what preceded t_{Planck}" is therefore not a neutral position. It is the admission of an emergence whose fundamental level has not yet been specified. ΛCDM is not non-emergent — it is emergent without specifying its own projection.

Our main theorem then applies: once the implicit projection is acknowledged, it must be non-injective. The non-injectivity that Cosmochrony makes explicit is not a feature of a particular model; it is a structural necessity that any complete successor to ΛCDM must satisfy.

Key message. A fully injective emergent description is a contradiction in terms: if Π were injective, the effective level would be isomorphic to the fundamental level, and there would be no emergence — only a relabelling.

2 Framework and Definitions

Let Ω be a space of fundamental configurations and $\mathcal{O}$ a space of observable states. We consider a surjective map $\Pi : \Omega \to \mathcal{O}$.

Definition 1 (Observables defined by Π)**.** *The observables of $\mathcal{O}$ are* defined by Π *if every structure on $\mathcal{O}$ — topology, algebraic relations, dynamical laws, or any other physically relevant structure — is entirely induced from Ω via Π, with no independent attribute assigned to $\mathcal{O}$ by an external postulate. Two configurations $\omega, \omega' \in \Omega$ are observationally indiscernible if and only if* $\Pi(\omega) = \Pi(\omega')$.

This definition is not an additional assumption: it is the formal translation of what is meant by a *purely emergent* description. $\mathcal{O}$ has no autonomous content; it is exactly what Π transports from Ω. Any theory that assigns independent postulated attributes to $\mathcal{O}$ is, by that very act, a two-level theory rather than an emergent one. Definition 1 therefore does not restrict the class of emergent theories; it *defines* it. Frameworks where $\mathcal{O}$ carries independent structure — such as standard quantum mechanics or general relativity, which directly postulate a Hilbert space or a spacetime manifold — lie outside this class by construction; see Section 6.

Definition 2 (Effectively non-trivial description)**.** *The description is* effectively non-trivial *if:*

(E1) ***Distinguishability:*** Π *is not constant — there exist* $\omega_1, \omega_2 \in \Omega$ *such that* $\Pi(\omega_1) \neq \Pi(\omega_2)$.

(E3) ***Structural non-isomorphism:*** $\mathcal{O} \ncong \Omega$ *as structured spaces, where "structured space" refers to the full set of physically relevant structures induced by* Π *(topology, algebraic relations, observables, or dynamical laws).*

Remark 1. *Condition (E3) is the formal statement that the effective level is genuinely distinct from the fundamental level. An isomorphism $\mathcal{O} \cong \Omega$ would mean that $\mathcal{O}$ is merely a relabelling of Ω, not an emergent structure. Note that (E2) — the statement that Π is non-injective — is deliberately absent from Definition 2: including it would make the main theorem circular.*

We also recall the standard projection entropy associated with Π. Given a non-degenerate measure μ on Ω, define $\nu(o) = \mu(\Pi^{-1}(o))$ for each $o \in \mathcal{O}$. The *projection entropy* is

$$S_\Pi \equiv -\sum_{o \in \mathcal{O}} \nu(o) \log \nu(o). \tag{1}$$

$S_\Pi = 0$ if and only if Π is (almost everywhere) injective; $S_\Pi > 0$ if and only if at least one fibre $\Pi^{-1}(o)$ has measure greater than that of a single point [2, 1].

3 Main Result

The proof of Theorem 2 rests entirely on the following lemma, which we state and prove separately to make the logical hinge explicit.

Lemma 1 (Injective projection is an isomorphism)**.** *Let $\Pi : \Omega \to \mathcal{O}$ be surjective and injective, with observables defined by Π in the sense of Definition 1. Then Π is a structural isomorphism $\mathcal{O} \cong \Omega$.*

Proof. Since Π is surjective and injective, it is bijective. Since all structure on $\mathcal{O}$ is induced from Ω by Π (Definition 1), a bijection that transports all structure is a structure-preserving isomorphism. Hence $\mathcal{O} \cong \Omega$. □

Remark 2 (Why this lemma is the pivot)**.** *Lemma 1 is the single logical hinge of the entire paper. Its content is deceptively simple: if nothing is lost under Π (injectivity) and everything in $\mathcal{O}$ comes from Ω (Definition 1), then $\mathcal{O}$ and Ω are the same structured space under different names. There is no emergence, only relabelling. Condition (E3) of Definition 2 forbids precisely this — and thereby forces non-injectivity.*

Theorem 2 (Compression necessity)**.** *Let $\Pi : \Omega \to \mathcal{O}$ be surjective, with observables defined by Π in the sense of Definition 1 and description effectively non-trivial in the sense of Definition 2. Then Π is necessarily non-injective, and for any non-degenerate measure μ on Ω,*

$$S_\Pi > 0.$$

Proof. Suppose for contradiction that Π is injective. By Lemma 1, $\mathcal{O} \cong \Omega$ as structured spaces. This contradicts condition (E3). Therefore Π is not injective.

Since Π is not injective, there exists $o \in \mathcal{O}$ such that $|\Pi^{-1}(o)| > 1$. For any non-degenerate measure μ, the weight $\nu(o) \in (0, 1)$ for this o, and $-\nu(o) \log \nu(o) > 0$. Hence $S_\Pi > 0$. □

4 Gauge Structure and Co-extensivity

Proposition 3 (Gauge requires non-injectivity)**.** *In any framework where observables are defined by Π (Definition 1), any non-trivial gauge structure in $\mathcal{O}$ requires Π to be non-injective, and hence $S_\Pi > 0$.*

Proof. If Π were injective, the holonomy group of Π would be trivial ($G_\Pi = \{e\}$), every connection would be flat, and no non-trivial gauge field could emerge from the bundle structure of Π alone. Any non-trivial gauge field would then require an independent postulate at the level of $\mathcal{O}$, violating Definition 1. □

Remark 3 (On the converse)**.** *The converse of Proposition 3 does not hold in general: non-injectivity does not automatically produce gauge structure. For the redundancy encoded in non-trivial fibres to organise into a gauge symmetry, additional geometric hypotheses are required: the fibres $\Pi^{-1}(o)$ must form a principal bundle over $\mathcal{O}$ with a non-trivial structure group G_Π admitting a connection. This is a result supplementary to Theorem 2, not a consequence of it.*

Corollary 4 (Co-extensivity)**.** *In any projective emergent framework satisfying the hypotheses of Theorem 2,*

$$\textit{descriptive redundancy} \iff \Pi \textit{ non-injective} \iff S_\Pi > 0,$$

and any non-trivial gauge structure requires these properties:

$$\textit{non-trivial gauge} \implies \Pi \textit{ non-injective} \iff S_\Pi > 0.$$

Under additional geometric hypotheses on the fibre organisation of Π *(principal bundle structure, admissible connection, non-trivial holonomy), the redundancy may in turn manifest as a gauge structure.*

5 Three-Level Structure

The results of Sections 3 and 4 establish a hierarchy with three distinct levels of increasing specificity.

Universal level. Theorem 2 holds for any surjective Π satisfying Definitions 1 and 2. No dynamical, geometric, or physical hypothesis is required beyond the structural conditions on Π. The conclusion — Π non-injective, $S_\Pi > 0$ — is a structural constraint on any framework that takes emergence seriously.

Intermediate level. If the non-trivial fibres of Π are organised geometrically — forming a principal G_Π-bundle with admissible connection — then the projective redundancy can support a gauge structure with holonomy group G_Π. This is an additional geometric hypothesis, not a consequence of Theorem 2 alone.

Cosmochrony level. In the Cosmochrony relational framework [1], the intermediate hypothesis is explicitly realised: the projection $\Pi : \mathcal{C}_\chi \to \mathcal{M}$ defines a principal bundle with fibre S^3, yielding $G_\Pi \supseteq \mathrm{SU}(2) \times \mathrm{U}(1)$. The resulting gauge structure underlies the emergence of electroweak interactions, while the Born–Infeld saturation bound selects the unique admissible infrared dynamics [3]. At the spectral level, the parity involution $c \leftrightarrow q - c$ of the Weil representation is a discrete realisation of projective fibre duality, and the pair observable $\sigma_{\mathrm{pair}} = \sigma_c \cdot \sigma_{q-c}$ is the admissibility observable dictated by this non-injective structure [6, 7]. Bell-type correlations arise as the observable signature of $S_\Pi > 0$ [2].

Remark 4 (On the route from non-injectivity to SU(2))**.** *Theorem 2 guarantees the existence of non-trivial fibres but says nothing about their form. The identification of* G_Π *with* $\mathrm{SU}(2)$ *rather than* $\mathrm{SO}(3)$ *or another group is a separate result, established in Cosmochrony via a chain of constraints absent from the universal theorem.*

The route is the following. The Born–Infeld parity constraint [7] fixes the minimal fibre structure to the involution $c \leftrightarrow q - c$*. The requirement that the representation algebra be associative, combined with the Hurwitz classification of real normed division algebras (*$\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$*), selects* $\mathbb{H}$ *as the minimal associative non-commutative structure [8]. The admissible neutral sector then has exactly* $\dim_{\mathbb{R}} \mathrm{Im}\,\mathbb{H} = 3$ *independent directions — the content of* $\Sigma_c(n_3) = 3$ *in O23 [8] — and the groups realising this sector form the set* $\{Q_8, \mathrm{Dic}_n, 2O, 2I\}$*, all subgroups of* $\mathrm{SU}(2)$ *and not of* $\mathrm{SO}(3)$ *[1]. The exclusion of* $\mathrm{SO}(3)$ *is structural: quotienting the central* $\mathbb{Z}_2$ *of* $\mathrm{SU}(2)$ *destroys the* 4π*-periodicity required for spinorial holonomy. In the continuum limit, the chain* $Q_8 \hookrightarrow 2I \hookrightarrow \mathrm{LPS}_p \hookrightarrow \mathrm{SU}(2)$ *converges to* $\mathrm{SU}(2)$ *as the unique spectral attractor.*

These steps use Born–Infeld dynamics, Weil representations, and the Hurwitz theorem — none of which appears in Theorem 2. The universal theorem supplies the necessary condition (non-trivial fibres); Cosmochrony supplies the sufficient conditions that fix their geometry. This selection is not universal but model-dependent: other emergent frameworks may yield different fibre geometries and different structure groups. The two results are complementary and non-redundant.

6 Scope and Limits of the Result

Theorem 2 is a conditional result. Its conclusion holds whenever the hypotheses of Definitions 1 and 2 are satisfied. It is therefore essential to identify the class of theories to which it applies and the class to which it does not.

Theories outside the scope. Standard physical frameworks — quantum mechanics, general relativity, standard-model field theories — are not covered by Theorem 2. These theories posit their fundamental objects (Hilbert space, spacetime manifold, gauge fields) directly at the level $\mathcal{O}$, without deriving them from a more primitive space Ω via a projection. The question of non-injectivity does not arise in such frameworks: there is no projection Π to be injective or otherwise. The theorem is silent about them.

Theories within the scope. Theorem 2 applies to any framework where the effective description is genuinely derived from a finer-grained level via projection. This includes, in particular: renormalisation group (RG) approaches, where the projection $\Pi_\Lambda : \Omega_{\mathrm{UV}} \to \mathcal{O}_{\mathrm{IR}}$ integrates out modes above the scale Λ and is non-injective by construction; causal set models, where macroscopic geometry is an invariant of sets of micro-configurations and the mapping is many-to-one; spin-foam and spin-network models, where the effective geometrical observables arise from averaging over discrete micro-structures; and relational frameworks such as Cosmochrony [1], where the full observable structure is explicitly derived from a pre-geometric substrate via a projection Π.

In each of these cases, the theorem does not prove non-injectivity from nothing: it shows that the non-injectivity already implicit in the framework's own definition of "effective description" is a logical necessity, not a contingent modelling choice.

The key distinction. The result isolates a precise logical boundary:

Injectivity of Π *is equivalent to the absence of genuine emergence.*

If Π is injective, the effective level is isomorphic to the fundamental level: nothing is lost, nothing is emergent. If the effective level is genuinely distinct, something is necessarily lost, and that loss is precisely $S_\Pi > 0$.

7 Categorical and Informational Reformulations

Two reformulations of Theorem 2 strengthen its connections to existing mathematical frameworks.

Categorical form. Let **Fund** be the category whose objects are spaces of fundamental configurations and whose morphisms are dynamically compatible maps, and let **Obs** be the category of observable spaces. The projection Π defines a functor $F_\Pi : \mathbf{Fund} \to \mathbf{Obs}$. Recall that a functor is *faithful* if it distinguishes all morphisms that the source category distinguishes; faithfulness is the categorical analogue of injectivity.

Theorem 5 (Categorical form of compression necessity)**.** *Under the hypotheses of Theorem 2, the functor* $F_\Pi : \mathbf{Fund} \to \mathbf{Obs}$ *is necessarily non-faithful. Conversely, a faithful functor from* **Fund** *to a non-equivalent* **Obs** *cannot exist.*

Proof. If F_Π were faithful, the induced map on morphisms would be injective at every hom-set, which at the level of objects recovers the injectivity of Π. By Lemma 1, this forces $\mathcal{O} \cong \Omega$, i.e. $\mathbf{Obs} \simeq \mathbf{Fund}$, contradicting condition (E3). Hence F_Π cannot be faithful. The converse follows immediately: a faithful functor to a non-equivalent category is a contradiction in the same sense. $\square$

Remark 5. *Theorem 5 identifies genuine emergence with the failure of faithfulness in the categorical sense. Faithful functors preserve all distinctions of the source; non-faithful functors compress them. Emergence is precisely this compression, now stated as a categorical necessity.*

Informational version. Let p be a probability distribution on Ω and $q = \Pi_* p$ its pushforward on $\mathcal{O}$. The Kullback–Leibler divergence of p with respect to the lifted distribution $\Pi^* q$ measures the information lost under Π:

$$D_{\mathrm{KL}}(p \,\|\, \Pi^* q) \;=\; \sum_{\omega \in \Omega} p(\omega) \log \frac{p(\omega)}{q(\Pi(\omega))}. \tag{2}$$

If Π is injective, $p(\omega) = q(\Pi(\omega))$ for all ω and $D_{\mathrm{KL}} = 0$. Theorem 2 implies:

Corollary 6 (Informational lower bound)**.** *In any effectively non-trivial emergent framework, there exists a distribution* p *on* Ω *such that* $D_{\mathrm{KL}}(p \,\|\, \Pi^* q) > 0$*. The projection entropy* S_Π *provides a distribution-independent lower bound on the structural information loss.*

The projection entropy S_Π defined in Eq. (1) is the distribution-independent part of this loss: it captures the structural compression built into Π independently of any particular state. D_{KL} then adds the state-dependent contribution. Together, they quantify two distinct aspects of the same fundamental compression.

The informational reading also connects to the renormalisation group. The Wilsonian RG projection Π_Λ is non-injective by construction: the same low-energy observable corresponds to infinitely many UV completions. Corollary 6 says that any such effective field theory has $S_{\Pi_\Lambda} > 0$ and $D_{\mathrm{KL}} > 0$: the loss of UV information is not a limitation to be overcome but a structural property of the emergent description itself.

8 Discussion

The central result admits a concise reading:

> *Any genuinely emergent physical description necessarily involves information compression. A fully injective projection would reduce emergence to a mere reparametrisation.*

The hypotheses are minimal. We do not assume a metric, a Hilbert space, a Lagrangian, or any physical dynamics. We require only that the effective description be defined by projection (Definition 1) and genuinely distinct from the fundamental level (condition E3).

The connection to information theory is direct. The projection entropy S_Π is not an epistemic quantity: it does not measure ignorance about which $\omega \in \Omega$ is the true configuration. It measures an objective structural property of Π — the degree to which Ω distinctions are compressed out of $\mathcal{O}$. Theorem 2 establishes that this compression is not optional: in any genuine emergent framework, $S_\Pi > 0$ is a structural necessity, not a contingent feature.

The result also clarifies the status of gauge symmetry. Gauge invariance has long been understood as a redundancy of description. Proposition 3 and Remark 3 make this precise in the projective setting: redundancy is equivalent to non-injectivity, which is itself forced by the structure of genuine emergence. Gauge symmetry is therefore not a fundamental feature of nature but a necessary organisational consequence of any emergent description that is rich enough to carry a principal bundle structure.

Projective incompleteness of standard physics. The frameworks ΛCDM and the Standard Model describe the structure of $\mathcal{O}$ with extraordinary precision, but they specify neither the fundamental level Ω nor the projection $\Pi : \Omega \to \mathcal{O}$. Yet three independent mechanisms within these frameworks each imply the existence of a non-injective projection.

(i) Cosmological singularity. The metric $g_{\mu\nu}$ ceases to be defined at $t \lesssim t_{\text{Planck}}$, so the geometry is effective, not fundamental. An implicit Ω and Π must exist; by Theorem 2, Π is non-injective.

(ii) Renormalisation group. The Standard Model is a Wilsonian effective field theory with UV cutoff Λ. The projection $\Pi_\Lambda : \Omega_{\text{UV}} \to \mathcal{O}_{\text{IR}}$ is non-injective by construction: infinitely many UV completions yield the same infrared observables.

(iii) Quantum measurement. The projection $\Pi_{\text{meas}} : \mathcal{H} \to \mathcal{O}_{\text{class}}$ maps distinct quantum states to the same classical outcome. The measurement problem is, in the present language, the question of the structure of Π_{meas} and why it is non-injective.

These three mechanisms are independent of any hypothesis specific to Cosmochrony; they follow from the internal structure of standard physics alone. Their common conclusion is that ΛCDM and the Standard Model are necessarily projectively incomplete. The following table identifies what is described and what is absent.

Level	Standard physics	Missing
Observable space $\mathcal{O}$	fully described	—
Fundamental space Ω	absent	unknown structure
Projection Π	absent	law of projection
Fibres $\Pi^{-1}(o)$	absent	microscopic multiplicity

The quantities that standard physics cannot derive — the fine-structure constant α, the electron-to-proton mass ratio m_e/m_p, the CKM mixing angles — are precisely the invariants of Π that $\mathcal{O}$ alone cannot determine: they encode properties of the fibres $\Pi^{-1}(o)$ that are invisible to any description confined to the effective level.

The incompleteness is not empirical but structural:

Standard physics omits the map that defines its own observables.

This imposes a precise constraint on any candidate unification or quantum-gravity programme: a theory that remains within $\mathcal{O}$ — however sophisticated — does not complete the description; it extends it. Any theory that genuinely completes standard physics must specify Ω, Π, and the non-injective structure of Π.

The three reformulations — structural (Theorem 2), categorical (Theorem 5), and informational (Corollary 6) — are equivalent in content but address different audiences and connect to different existing bodies of work. Their convergence on the same conclusion strengthens the claim that non-injectivity is a universal structural feature of emergent descriptions, not an artefact of any particular formalism.

The core equivalence, within the present projective-emergent setting, can be stated in a single line:

$$\text{genuine emergence} \iff \text{information loss} \iff \Pi \text{ non-injective} \iff S_\Pi > 0. \tag{3}$$

The left-to-right direction is Theorem 2; the right-to-left direction follows from the fact that any non-injective Π whose fibres are non-trivial defines a genuinely distinct effective level by the identification of distinct configurations. A fully injective effective theory is not an emergent theory but a renaming of the fundamental level.

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